1. Clinical Question & Core Hypothesis

Clinical Question

Why do stromal-rich breast tumors progress/metastasize early despite abundant stromal microenvironment — i.e., what tumor-intrinsic programs within a high stromal-to-tumor ratio (TSR) niche drive EMT and early dissemination? 2) Tier 1 paths used: the GLS ↔ Cyclin D1 (CCND1) metabolic–proliferation axis and the EMT→metastatic spread linkage. I built on these because GLS links metabolism to cell-cycle control (CCND1), which can fuel EMT-associated invasive behavior that leads to progression. 3) Tier 2 evidence that supports the claim: CXCR4–GLS association and Wnt/β-catenin's role in reinforcing EMT/metastatic programs provide plausible upstream modulators of the GLS–CCND1 axis and a mechanism by which the stroma could promote metabolic reprogramming. 4) Endpoint choice: PFI (progression-free interval) was selected because the mechanistic chain explicitly concerns EMT-driven dissemination and early progression/metastatic events rather than long-term mortality; PFI most directly captures timing of progression. 5) Deliberate exclusions: I excluded immunotherapy response and MCL-specific context (cyclin D1 in lymphoma) from the core claim to keep focus on breast tumor progression mechanisms; CXCR4, Wnt/β-catenin and drug-transporter/drug-combination elements are relegated to secondary_modifiers to avoid exceeding the 2-gene limit and to keep the mechanistic question sharp?

Working Hypothesis

In breast tumors with a high stromal-to-tumor ratio (TSR), GLS-driven glutaminolysis sustains CCND1-mediated proliferation and reinforces EMT programs, creating an intrinsic tumor-intrinsic axis that enables early metastatic dissemination. Consequently, TSR-high tumors with concurrent high GLS and CCND1 expression will exhibit shorter progression-free interval (PFI), and this adverse outcome will be amplified when CXCR4 signaling or WNT/β-catenin activity is elevated, with potential modulation by expression of drug transporters.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	GLS, CCND1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GLS: GLS association and Wnt/β-catenin's role in reinforcing EMT/metastatic programs provide plausible upstream modulators of the GLS–CCND1 axis and a mechanism by which the stroma could promote metabolic reprogramming.

CCND1: CCND1 axis and a mechanism by which the stroma could promote metabolic reprogramming.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CXCR4	Supporting modifier from knowledge graph

Secondary	Wnt/beta-catenin signaling	Supporting modifier from knowledge graph
Secondary	Drug transporters (e.g., ABC family)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: in vitro models that recapitulate TSR-like stromal interactions using fibroblast-rich co-cultures;

Prediction 2: genetic (CRISPR-Cas9) or pharmacologic perturbations of GLS, CCND1, CXCR4, and WNT signaling;

Prediction 3: in vivo xenograft or PDX models incorporating stromal compartments to assess PFI/ metastatic timing; and

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CXCR4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Wnt/beta-catenin signaling: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.